| INCH-POUND |

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SUPERSEDING
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14 June 1985

MILITARY SPECIFICATION

PUMP UNIT, CENTRIFUGAL, SALT WATER, DIESEL-ENGINE-DRIVEN

'This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel-engine-driven, salt water pump units The pump units are mounted on a rigid steel skid mounting that forms a compact, self-contained, movable unit.

* 1.2 Classification, The pump units are of the following styles and sizes, as specified (see 6.2):

STYLES

Style A - Axially split, side suction.
Style B - Radially split, end suction,

SIZES

Size 3 - 3-inch discharge, 3- or 4-inch suction
Size 4 - 4-inch discharge, 4-inch suction.
Size 6 - 6-inch discharge, 6-inch suction.

|Beneficial comments (recommendations, additions, deletions) and any pertinent|
|data which may be of use in improving this document should be addressed to: |
|Commanding Officer (Code 156), Naval Construction Battalion Center, Port |
|Hueneme, CA 93043-5000, by using the self-addressed Standardization |
|Document Improvement Proposal (DD Form 1426) appearing at the end of this |
|document or by letter. |

AMSC N/A FSC 4320

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2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in tile solicitation (see 6.2)

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting, and Starting.

MILITARY

MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water; Packaging of.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1,2 Other Government Publications. The following other Government publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Title 29, CFR, Part 1910.95 - Occupational Noise Exposure.
Title 29, CFR, Part 1910.219 - Mechanical Power-Transmission Apparatus,

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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ENVIRONMENTAL PROTECTION AGENCY

Title 40, CFR, Part 86 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines" Certification and Test Procedures.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
ASME B 1.20.1 - Pipe Threads, General Purpose (Inch).
ASME B 1.20.3 - Dryseal Pipe Threads (Inch).
ASME B 151 - Safety Standard for Mechanical Power Transmission Apparatus.
ASME B 16.1 - Cast Iron Pipe Flanges and Flanged Fittings
ASME B 16.5 - Pipe Flanges and Flanged Fittings
ASME B 16.24 - Bronze Pipe Flanges and Flanged Fittings, Class 150 and 300.
ASME B 40.1 - Gauges - Pressure Indicating Dial Type - Elastic Element.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017,)

ASTM

ASTM A 582 - Free-Machining Stainless and Heat-Resisting Steel Bars , Hot-Rolled or Cold-Finished.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Standards of Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Ave., Cleveland, OH 44107.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

SAE J534 - Lubrication Fittings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc , 400 Commonwealth Drive , Warrendale, PA 15096)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services)

2.3 Order of Precedence . In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document. however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description The pump unit style and size shall be as specified (see 1.2 and 6.2). The pump unit shall be single- or multi-stage, single- or double-suction. Suction and discharge connections shall be flanged.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Standard commercial product , The pump unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer' s standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the pump unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

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3.5 Design and construction. The pump unit shall be designed and constructed to withstand the strains, jars, vibrations, and other conditions associated with shipping, storage, and installation; and shall be complete so that when installed, it can be used for any operation for its intended purpose, The pump unit and accessories shall be designed and constructed to prevent conditions hazardous to personnel or deleterious to -equipment, shall withstand the hard usage encountered in military service or shore establishments, and shall permit easy and readily accessibility for replacement of accessories, maintenance, service, and adjustments in the field. All threaded parts shall be in the inch system and shall conform to ASME B 1.1, ASME B 1.20.1, and ASME B 1.20.3, as applicable.

3.5.1 Vibration. Each pump unit shall be so designed and balanced that any natural frequency causing excessive torsional or resonance vibration shall occur at not less than 25 percent above the maximum governed speed of the engine.

3.5.2 safety. All mechanical and electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The pump unit shall comply with Title 29, CFR, Part 1910.95 for noise and Part 1910.219 for guarding,

3.5.3 Pump casing. The pump casing shall be of good commercial bronze, 316 stainless steel, or cast-iron, bronze-fitted, as specified (see 6.2). The casing shall be capable of withstanding a hydrostatic test pressure equal to 1.5 times the shutoff head at the maximum brake horsepower (hp) of the engine at the applicable pump speed. For style A pump units, the casing shall be axially split along the centerline of the pump shaft and shall have inlet and outlet flanged connections cast integral with the lower casing. Style B pump units shall be furnished with radially split casings and end suction connection. The casing shall have necessary tapped openings for air venting, priming, draining, and for suction and discharge gages. A brass, stainless steel, or bronze petcock shall be furnished for air venting purposes. Drain openings in the volute, intake, elbows, reducers, or other passages capable of retaining trapped water shall be located in the low point of such passages when the pump unit is in a level position. A threaded plug or petcock of brass, stainless steel, or bronze shall be installed in each drain. When pump unit designs are such as to make drains inaccessible, short drain pipes of brass, stainless steel, or copper shall be installed. Petcocks, plugs, and drain pipes will normally be of the same material as the pump casing. Use of other materials cited herein may be acceptable only if no deterioration of parts will occur due to galvanic corrosion.

3.5.4 Suction and discharge connections. Suction and discharge connection shall be of the nominal sizes specified in 1.2, and shall be flanged. Threaded companion flanges shall be furnished on both connections of each pump unit. Unless otherwise specified (see 6.2), style B pump units shall also be furnished with suitable elbows to place the suction and discharge connections on opposite sides of the pump to facilitate installations in a horizontal run of piping, The elbows shall be located so that the centerline through the pump and engine is perpendicular to the horizontal run. Discharge and suction connections shall be suitably identified by appropriate words or arrows which shall be cast integrally with the casing, traced by welding bead, cut into the casing, or inscribed on a firmly attached nonferrous plate. A suitable flanged reducer shall be incorporated with double intake pumps, as required to eliminate

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hydraulic imbalance caused by unequal flow to the intakes of the impeller All bronze pumps shall have bronze flanges conforming the ASME B 16.24, class 150. Cast iron pumps and threaded iron companion flanges bolted to pump suction and discharge connections shall conform to ASME B 16.1, less 125 Stainless steel pumps and threaded companion flanges bolted to pump suction and discharge connections shall conform TO ASME B 16 5, class 150

3.5.5 Impeller. The impeller shall be either an open or enclosed type and shall be of 316 stainless steel or bronze cast in one piece in accordance with the manufacturer's standard practice. The impeller shall be dynamically balanced, keyed and held firmly in place to prevent lateral movement. The impeller shall be capable of passing an ingested solid equivalent to a 5/16-inch sphere .

3.5.6 Wearing rings Renewable bronze or carbon-graphite wearing rings shall be provided for the impeller or casing, or both, as required, to facilitate elimination of excessive leakage from the discharge to the suet-ion sick of the pump. Wear rings shall be securely locked to prevent lateral and rotational movement

3.5.7 Check valves. A check valve shall be provided at the discharge outlet of the pump for sealing the pump during priming unless a self-priming pump is furnished. When a swing check valve with attached external lever is used as a means of applying manual pressure to improve the valve seal, the valve shall have a suitably located, firmly attached brass or copper plate that indicates the proper lever operation

3.5.8 Pump shaft. The pump shaft shall be of corrosion-resistant steel alloy, not less than ASTM A 582, alloy 303, or a nickel-copper alloy (Ni 67.0 Cu 30.0) or equal having physical properties consistent with good commercial practice for the intended use. Pump shafts shall require sleeves in the stuffing boxes to eliminate wear of the shaft if fibrous packing is used. Alloy steel shafts shall be furnished with 17-4 PH sleeves. Nickel-copper shafts shall be furnished with nickel-copper-aluminum alloy sleeves. The pump shaft shall be coupled to the engine crankshaft extension with a flexible coupling of a type normally furnished by the manufacturer for the size pump unit specified herein. A safety guard of solid or expanded sheet metal in accordance with ASME B 15.1 shall be securely installed over the coupling and the shaft.

3.5.9 Seals. The pump casing shall be sealed against leakage where the pump shaft passes through the casing by the use of either fibrous-packed stuffing boxes or mechanical seals at the option of the manufacturer. Fibrous-packed stuffing boxes shall be fitted with split bronze or stainless steel glands. The portion of the shaft that passes through the fibrous-packed stuffing box shall be grease-lubricated. Lubrication and sealing of the shaft in the packing by use of salt water from the pump casing will not be permitted. When mechanical seals are furnished, removable sleeves will not be required in the stuffing box, as specified for alloy steel and nickel copper metal shafts (see 3.5.8).

3.5.10 Bearing assemblies. Bearings shall be separate from the stuffing boxes and shall be of the grease-lubricated, antifriction type, one set on each side of the casing for style A pump units, and two sets of bearings placed well apart on the same side for style B pump units. Bearing housings shall be located so as not to interfere with the prepacking and maintenance of the

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stuffing boxes and shall be completely sealed against the entrance of grit, water , and other foreign materials. Bearings shall be protected, as required, by rigid liquid deflectors mounted and locked on the shaft between the stuffing box and the bearing housing. One bearing shall be designed to take both thrust and radial loads, the other may be designed for radial loads only.

3.6 Performance. Unless otherwise specified (see 6.2), the pump units shall meet the following capacities, using salt water at 68 degrees Fahrenheit (°F) or at room temperature for which proper specific weight corrections have been applied. The performance requirements shall be met with ambient atmospheric conditions corrected to standard sea level conditions of temperature and pressure.

a. Size 3 - 300 gallons per minute (gpm) with a total dynamic head of 190 feet including a 15-foot suction lift.
b> Size 4 - 500 gpm with a total dynamic head of 152 feet including a 15-foot suction lift,
c, Size 6 - 500 gpm with a total dynamic head of 231 feet including a 5-foot suction lift.

The performance shall be met with the engine controls set to produce continuous duty rating. To preclude the use of an overpowered pump unit, the test shall be met, but not exceeded by more than 20 percent while the engine speed controls are set at an adjustment producing not less than 60 percent of the net continuous brake hp rating. Pump performance shall be measured at these approximate percentages of rated capacity: O, 25, 50, 75, 100, and 125. When these points are plotted, the pump performance curve shall produce a smoothly rising head-capacity tune on which heads at the several discharge rates are operating with the speed control adjusted as specified herein. The pump efficiencies shall equal or exceed those shown in the figures. The design of the pump units shall be such that the pumps shall not cavitate or tend to cavitate while pumping salt water at 85°F maximum with a 15-foot lift measured at the pump suction.

3.7 Engine. Unless otherwise specified (see 6.2), the engine shall be a standard commercial liquid- or air-cooled diesel engine. When specified (see 6.2), the engine shall be a liquid-cooled gasoline engine. The engine shall have a continuous hp rating sufficient to operate the pump units co meet the performance requirements of 3.6 and shall comply with Title 40, CFR 86 or the applicable CFR in effect at the time of manufacture pertaining to the control of air pollution. Gasoline engines, when specified, shall produce the required performance equally as well on unleaded or leaded gasoline. All engine accessories supplied commercially as standard equipment including instruments and gages shall be furnished. The engine shall be furnished complete with, but not limited to, the following accessories:

a. Governor.
b. Air cleaner with precleaned and restriction indicator gage.

3.7.1 Starting systems. The engine shall be equipped with a 12-volt (V) electric starting system, The starting system for the diesel engine may be either electric glow plug or ether primer. The diesel engine shall start at a temperature of -20°F. When ether priming system is required, it shall be of the measured-shot type, with storage capacity of not less than 12 fluid ounces. The

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electric starting system for the gasoline engine, when specified (see 3 7), shall include an auxiliary manually operated crank. The gasoline engine shall be equipped with a manual or automatic, ignition-spark retard device for safety, and ease of hand-cranking. A suitable, readily visible instruction plate shall be furnished and shall outline the step-by-step procedure that ensures safety and ease of starting the engine. The plate shall conform to 3.11

3.7.2 Exhaust system. An exhaust system and muffler shall be furnished and shall be the manufacturer's standard design, except for modifications that may be necessary to supply without damage to the engine, muffler, or exhaust systems , sufficient volume and pressure of exhaust gas to operate a commercial type, ejector pump primer, when such a primer is furnished (see 3.8.3).

3.7.3 Batteries . Batteries shall be furnished. Batteries shall conform to W-B-131 and shall have a reserve capacity of not less than 385 amperes. Batteries shall be shipped in a charged and dry condition. Battery cables shall be the standard automotive type.

3.7.4 Charging system. The engine shall be equipped with alternator and regulator rated at 12V direct current. The alternator shall deliver sufficient current to charge the batteries, and when required (see 3.8,4) , supply the power requirements of the auxiliary lighting system.

3.7.5 Control system. The control system shall be furnished, equipped with the following devices:

a. Gage, pressure, lubricating oil
b. Voltmeter, battery-alternator.
c. Gage, cooling liquid temperature, unless air-cooled.
d. Starter switch.
e. Suitable waterproof auxiliary power connector if auxiliary lighting is required (see 3.8.4).
f. Glow plugs or ether primer starting system.
g. Suitably located incandescent bulbs for indirect lighting of instrument panel mounted gages and controls.
h. Fuel gage.
i. Engine hour meter that registers engine operating time in hours, capable of registering at least 9,999.9.

3.8 Components and accessories.

3.8.1 Pressure gages. A compound gage reading from O to 30 inches of mercury vacuum and O to 150 pound-force per square inch (psi) shall be furnished for the pump suction. A pressure gage with a scale range of at least twice the total dynamic head specified in 3.8 shall be furnished for the pump discharge. Both gages shall be furnished with pulsation dampers and gage cocks. Dials shall be not less than 2-1/2 inches in diameter. Gages shall be of the Bourdon type and shall conform to the requirements of ASME B 40.1.

3.8.2 Suction strainer . A suction strainer having the same nominal pipe size as the inlet connection shall be furnished with each pump unit. The strainer shall have a net area of at least four times that of the suction pipe but shall not allow solids to pass larger than specified in 3.5.5

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3.8.3 Auxiliary priming systern Unless otherwise specified (see 6.2), a dual component, auxiliary priming system shall be furnished. When specified (see 6.2), the pump shall be self-priming. For the auxiliary priming system, one component of the system shall be a commercially available hand pump capable of filling the centrifugal pump and suction line with priming water The other component shall be manually controlled, exhaust ejector, or a manually controlled, suction device that makes use of the vacuum created in the intake manifold of the engine to prime the pump. Automatic-means , by float or otherwise , shall be provided to prevent entrance of water into the intake manifold during primer operation. Means shall be provided for isolating the components from the centrifugal pump and for draining each component of the system so that drainage goes back into the volute or other areas after operation. A readily visible brass or copper plate securely fastened in the vicinity of each of the two priming devices shall outline, step by step, the operating procedure for each. The plate shall include a warning against running the engine to accomplish priming of a dry pump by use of the ejector or vacuum system except for repriming purposes in which the residual water of the volute can act as a lubricant. If the volute is dry, the pump casing shall be filled with water prior to using the engine-driven priming system. When an ejector primer is furnished, the instructions shall specify that the ejector shall be dismantled and cleaned each 200 hours of engine operation

3.8.4 Auxiliary lighting system. When specified (see 6.2), an auxiliary lighting system for night operation shall be provided. The system shall consist of two separate lighting units consisting of bulb, lens-enclosed reflector, spring-actuated clamp or magnetic clamp for reflector, 20-foot cord with jack-plug, or terminal leads for connection to the instrument panel. The cord shall be in a accordance with NFPA 70, type SO, for use in damp places, and comprising two conductors of a size sufficiently rated to carry current twice the lamp amperage. The bulbs shall be of 60-watt, 12Vdc, and shall be of a standard waterproof bayonet-socket type.

3.8.5 Firehose connections. When specified (see 6.2), a flanged T, Y, or 45-degree lateral connection shall be made to the discharge side of the pump. The size 3 and 4 branch connections shall utilize a globe or gate valve of suitable capacity terminating in one firehose male nipple, provided with a captive-chained cap, with both nipple and cap having 2-1/2-inch hose thread conforming to the following:

a. Threads per inch - 7-1/2
b. Pitch - 0.13333 inch
c. Depth of thread - 0.08660 inch
d. Pitch diameter - Maximum - 2.9820 inch
Minimum - 2.9660 inch
Tolerance - 0.0160 inch

Both nipples and caps shall be capable of repeated use. The main line of the T, Y, or 45-degree lateral connection shall utilize a flanged gate valve of suitable capacity connected to the discharge side. The firehose connections shall remain attached to be instantly available for emergency use.

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3 8.6 Skid mounting The pump and engine shall be securely mounted on a steel skid mounting, properly braced and reinforced by cross members to adequately support the fully assembled pump unit and hold the pump and engine in proper alignment, particularly during times when they are towed along the ground or handled during shipment. Each runner of the skid mounting shall have both ends upturned in a manner to facilitate skidding. Provision shall be made for bolting the mounting to a permanent foundation by uniformly spaced holes in the skids , by non-projecting lugs, or by other means.

3.8.7 Lifting and tiedowm attachments The pump unit shall be equipped with lifting and tiedown attachments conforming to MIL-STD-209, type II or type III. A nonferrous transportation plate shall be provided and mechanically attached to the pump unit. Transportation places shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling. A silhouette of the pump unit furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for tying down the pump unit on the carrier when shipped.

3.9 Interchange ability All pump units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories , and spare parts.

3.10 Identification plate. An identification plate will be furnished by the contracting officer for each pump unit. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each pump unit in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. Nomenclature shall be “PUMP UNIT, _____ GPM, (GED or DED)" as applicable.

3.11 Instruction plates. The pump unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.12 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer’s standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The pump unit shall be lubricated as specified in section 5.

3.13 Cleaning, treatment, and painting . Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other interfering substances. As soon as practicable after cleaning and before any corrosion product or other

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contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 roils over the entire surface The paint shall be free from runs, sags , orange peel, or other defects. The color of the finish coat shall be in accordance with FED-STD-595 and the color number shall be green, No. 14064. The end item, allied equipment, and attachments shall be the same color.

3.14 Workmanship, The quality of workmanship shall be such as to produce pump units that are in accordance with the requirements of this specification and to ensure proper functioning of all parts of the pump unit,

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be delecterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.14.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Castings, All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.14.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibiliy for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract . Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection . The first article inspection shall be performed on one pump unit when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and when specified (see 6.2), the packaging inspection of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. In the event the contractor desires to deliver the test model as a contract item, it shall be delivered as the last item on the contract only after the contractor, at his own cost and expense , shall have completely cleaned, reconditioned, and overhauled the pump unit, making such replacements and modifications thereto as are required to make the pump unit acceptable as a contract item.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.2, and the packaging inspection of 4.5.

4 3 Examination. Each pump unit shall be examined for compliance with the requirements specified in section 3 of this document, Any redesign or modification of the contractor's standard product to comply with specified requirements , or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.4 Tests.

4.4,1 First article tests. Performance tests shall be conducted In accordance with the test standards for centrifugal pumps of the Standards of Hydraulic Institute. The first article shall receive the tests of 4.4.1.1, 4.4.1.2, 4.4.1.3, and 4.4.1.4. Failure to pass any test shall constitute cause for rejection.

4.4.1.1 Operational test. The pump unit shall be run-in for a period of not less than ten minutes, at operating speed, to satisfy the operability of the pump unit. Any evidence of failure in alignment of the pump and driver, operability, or evidence or excessive temperature, vibration, or noise shall constitute failure of the operational test.

4.4.1.2 Performance test. The pump unit shall be tested to meet the capacity and head requirements of 3.6,1. The pump shall be operated continuously for four hours. Capacity, head, smoothness of operation, effectiveness of stuffing box or mechanical seals, and bearing temperatures shall be checked during this test. Failure of the pump unit to meet any capacity-head requirements shall constitute failure of the test.

4.4.1.3 Vibration. The pump unit shall be tested to meet the requirements of 3.6.2. Unless otherwise specified (see 6.2), the vibration limits and test procedures shall be in accordance with the Standards of Hydraulic Institute.

4.4.1.4 Hydrostatic pressure test. All pump casings produced shall be tested to verify compliance with 3.6.3. All pump casings shall be subjected to a hydrostatic pressure equal to 1.5 times the pump discharge pressure or the total dynamic head, whichever is greater, for a period of not less than five minutes . Minor leakage through the mechanical seal shall not be cause for rejection. Any damage or permanent deformation to the pump casing or leakage in the casing or through the casing gasket shall constitute failure of the test.

4.4.2 Quality conformance tests. Each production pump unit shall be operated and tested as specified in 4.4.1.1 and 4.4.1.4.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-P-10603. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-10603 with the level of presentation and the level of packing as specified (see 6.2).

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The pump units are intended for applications within their capabilities such as the following

a Jetting in pile-driving operations.
b Standby and emergency fire protection.
c General utility service in temporary and semi-permanent locations

In all uses stated, the pump unit will not be expected to pass ingested solids of greater size than a 5/16-inch sphere.

* 6.1.1 Selection factors.

a. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit. elements, including terminal and circuit connections, when the pump unit is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated
b. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirements and test limits as specified in SAE J551,

6.2 Acquistion requirements. Acquisition documents should specify the following:

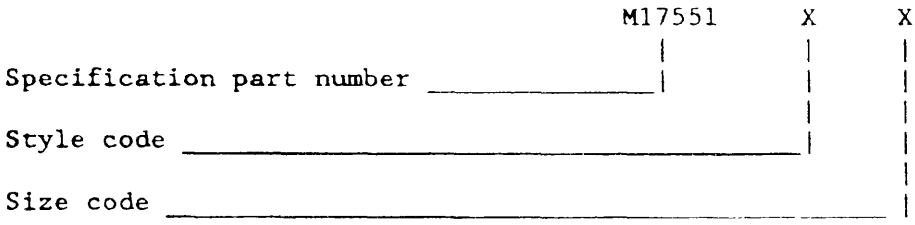
a. Title, number, and date of this specification.
b. Style and size of pump unit required (see 1.2 and 3.1)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required for inspection and approval (see 3.2 and 4.2.1).
e. Whether pump casing shall be bronze, 316 stainless steel or cast iron, bronze-fitted (see 3.5.3).
f. When fittings for providing connections on opposite sides of style B, end suction are not required (see 3.5.4).
g. When capacities, head or suction lift is to be other than as specified (see 3.6).
h. When a gasoline engine is required (see 3.7).
i. When starting system is to be other than as specified (see 3.7.1)
j. When an auxiliary priming system is not required or the priming system is to be self-priming (see 3.8.3).
k. When an auxiliary lighting system is required (see 3.8 4)
1. When firehose connection(s) are required (see 3.8.5).
m. When lubrication is other than as specified (see 3,12).
n. When a preproduction pack inspection is required and the time frame required for submission (see 4.2.1 and 4.5).
o. When vibration tests are to be other than as specified (see 4 4.1 3).
p. Level of preservation and level of packing required (see 5.1)

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6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one pump unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Part or identifying number (PIN). The PIN to be used for pump units applied to this specification are created as follows



6.5.1 Style to code identifier. The style of the pump unit is identified by a single digit as shown in table I.

TABLE I Type to code identifier

Style	Code
Style A	A
Style B	B

6.5.2 Size to code identifier. The size of the pump unit is identified by a single digit as shown in table II.

TABLE II. Size to code identifier.

Size	Code number
Size 3	3
Size 4	4
Size 6	6

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* 6.6 Subject term (key word) listing.

Axially split casing
Radially split casing
Skid base mounted
Single stage
Multi-stage

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections , deletions) from the previous issue were made This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity-
Navy - YD

(Project 4320-N333)

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DOCUMENT TITLE
PUMP UNIT, CENTRIFUGAL, SALT-WATER, DIESEL-ENGINE-DRIVEN

NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ANNE L. LYONS	b. TELEPHONE (Include Area Code) (1) Commercial 805 982-5761	(2) AUTOVON 551-5761
c. ADDRESS (Include Zip Code) NSBC CESO (CODE 1564A) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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